External Higgs Signals and Latent-Space Model Selection

Comparing a balanced representation model and an anomaly-oriented model against external Higgs-like signals, plus an exploratory VICReg-style check

Estefania Moreno León on behalf of the NextGen WP3 OpenLab -

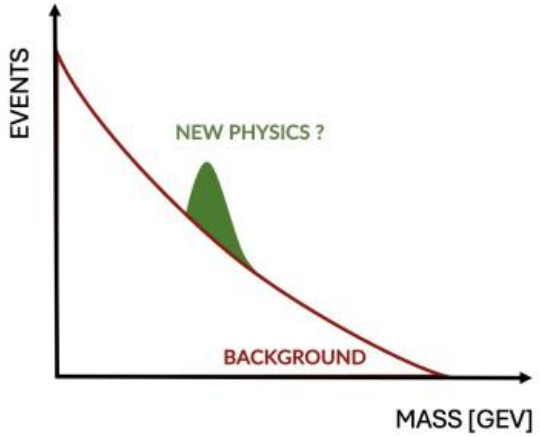
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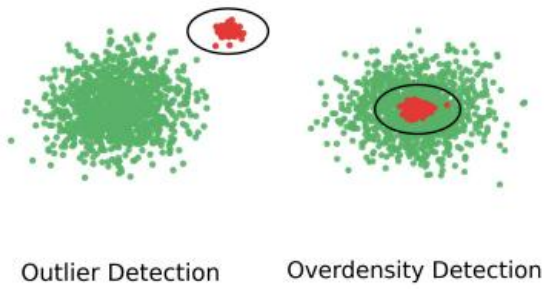
Anomaly detection at L1, not only classification

- This project is about **anomaly detection** at L1 trigger level, **not only classification**.

- The goal is to **learn a latent representation** of L1-only event information that can help identify rare or unseen signals.

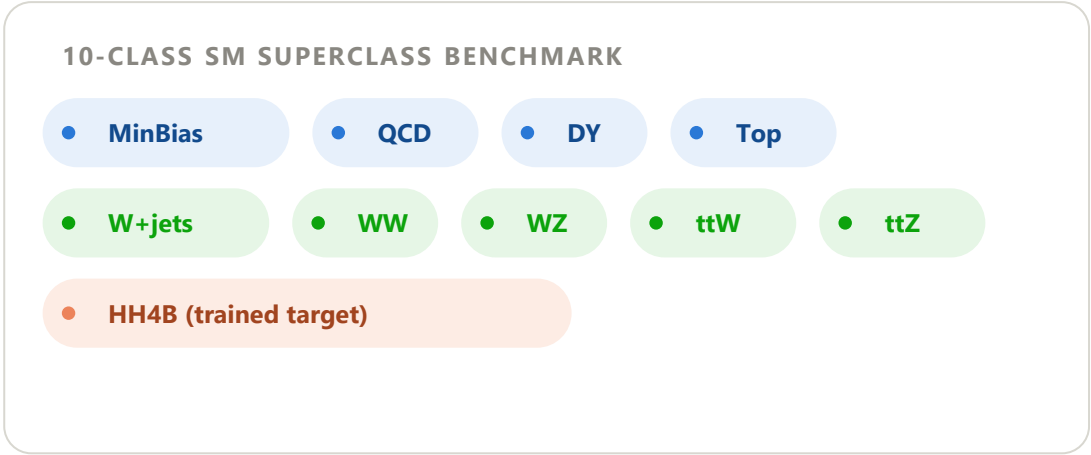
- Current focus: understand how HH4B and external Higgs-like samples behave in latent space.

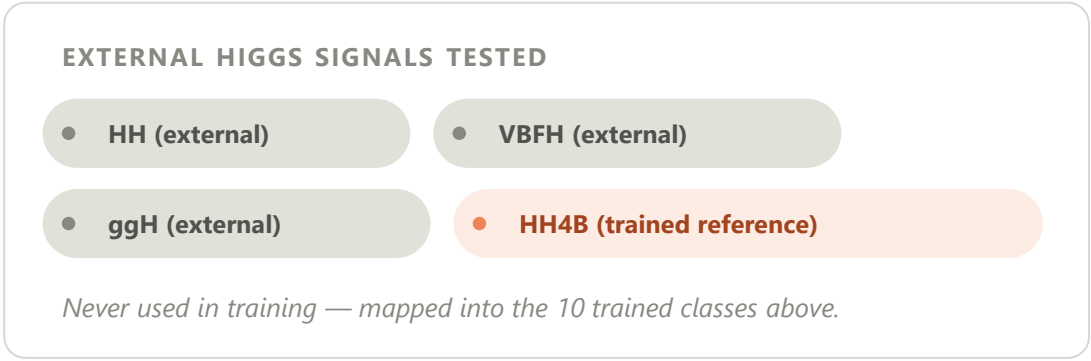




Five new pieces of work since Presentation 3

- **Broader label set:** 5-class HH4B setup → 10-class SM superclass setup.
- **Objective sweep:** evaluated 6 objective/model configurations.
- **Capacity sweep:** re-ran the best balanced configuration at latent_dim = 32, 64, 128.
- **External signal evaluation (new):** added HH, VBFH, ggH, evaluated alongside the trained HH4B class.
- **Exploratory objective (new):** added VICReg-style representation-learning tests.





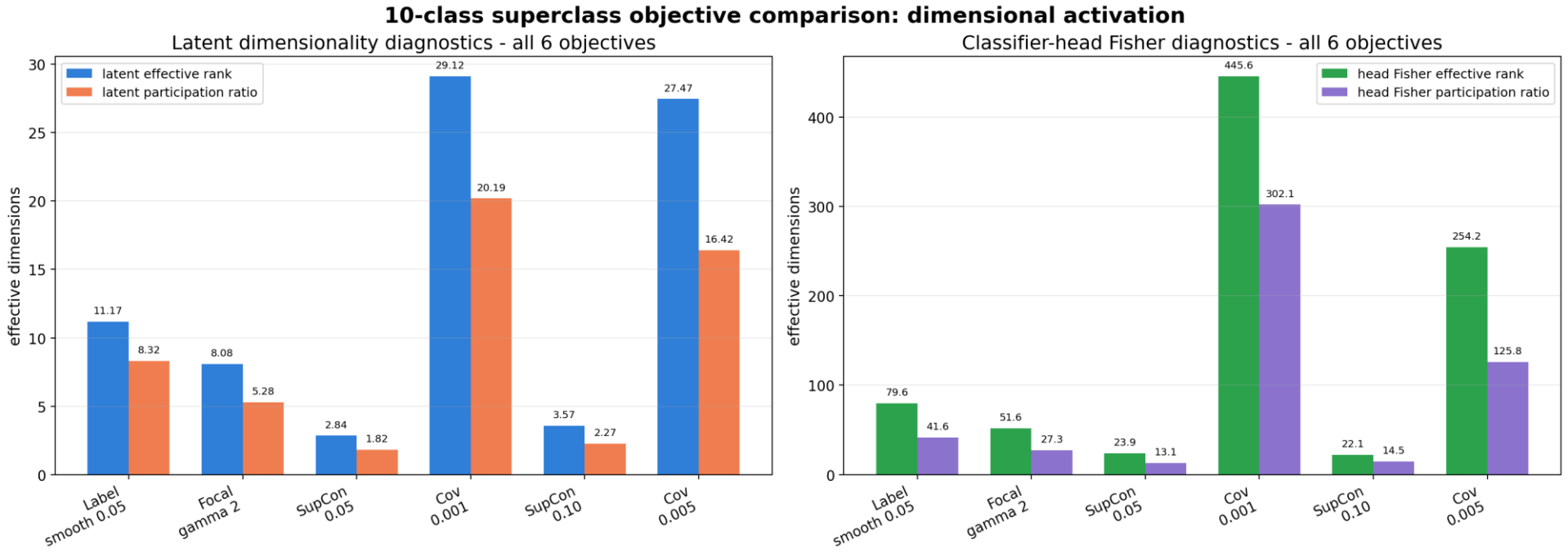
Objective choice, not just capacity, drives the spread

All at latent_dim=64, ff_dim=128, heads=8, 2 blocks, mean pooling, linear head, 100k events/class

CONFIGURATION	ACCURACY	MACRO F1	HH4B F1	EVAL-QUALITY	EFF. RANK	HEAD-FISHER RANK
Label smoothing 0.05 + var	0.4747	0.4677	0.5613	0.00004	11.17	79.63
Focal γ=2 + var	0.4600	0.4543	0.5567	0.1476	8.08	51.63
CE + SupCon 0.05 + var ★ balanced	**0.4602**	**0.4560**	**0.5612**	**0.2207**	**2.84**	**23.87**
CE + var + Cov 0.001 ★ anomaly-oriented	**0.4556**	**0.4468**	**0.5566**	**0.2294**	**29.12**	**445.55**
CE + SupCon 0.10 + var	0.4662	0.4580	0.5609	0.1937	3.57	22.14
CE + var + Cov 0.005	0.4716	0.4644	0.5599	0.1318	27.47	254.24

⚠ **A plain "CE + variance only" 10-class run is not available in this comparison — all six rows add a further regularizer on top of CE + variance.**

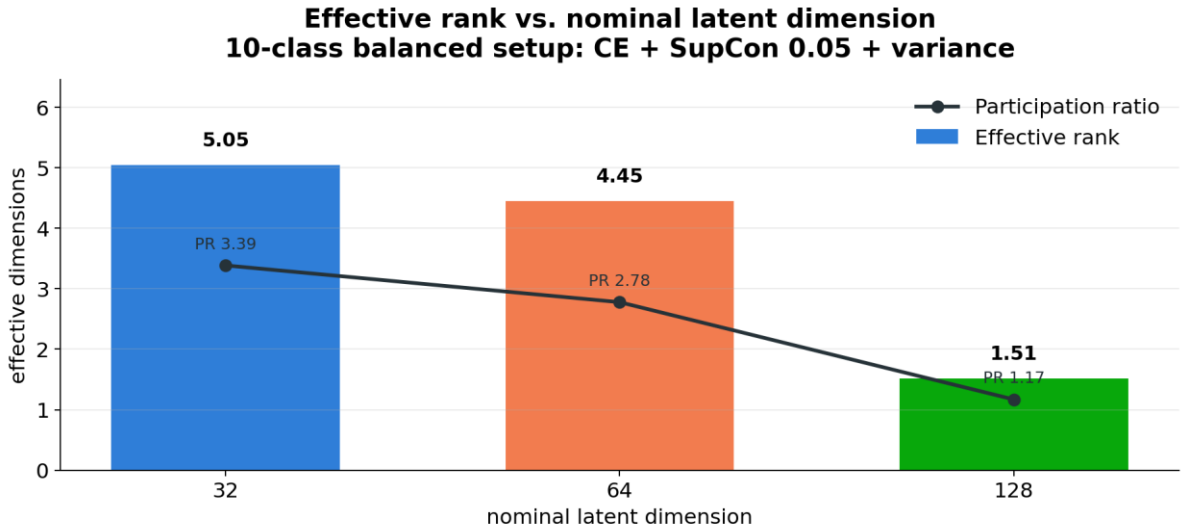
Testing six CrossEntropy-based objective variants



- **Effective rank** measures how many latent directions are actively used by the representation.
- **Head-Fisher effective rank** measures how many classifier-head directions are important for the task.
- All six configurations share the same latent_dim=64 — yet effective rank ranges from 2.84 to 29.12, roughly a 10x spread, from objective choice alone.

More capacity did not clearly improve the representation

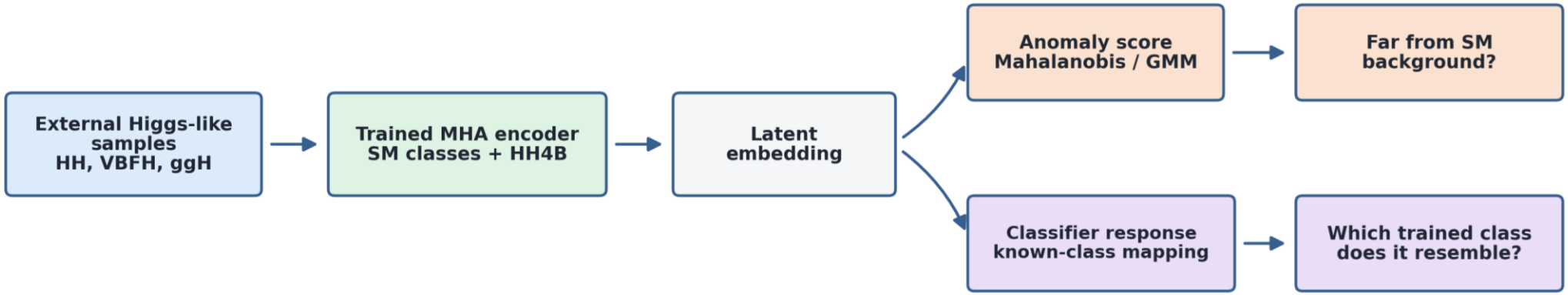
CE + SupCon 0.05 + variance · 100k events/class · same architecture otherwise



LATENT_DIM	ACCURACY	MACRO F1	HH4B F1	EFF. RANK	PARTIC. RATIO	EVAL-QUALITY
32	0.4687	0.4634	0.5639	5.049	3.386	0.1594
64	0.4734	0.4695	0.5593	4.448	2.778	0.00004
128	0.4675	0.4613	0.5615	1.511	1.169	0.2253

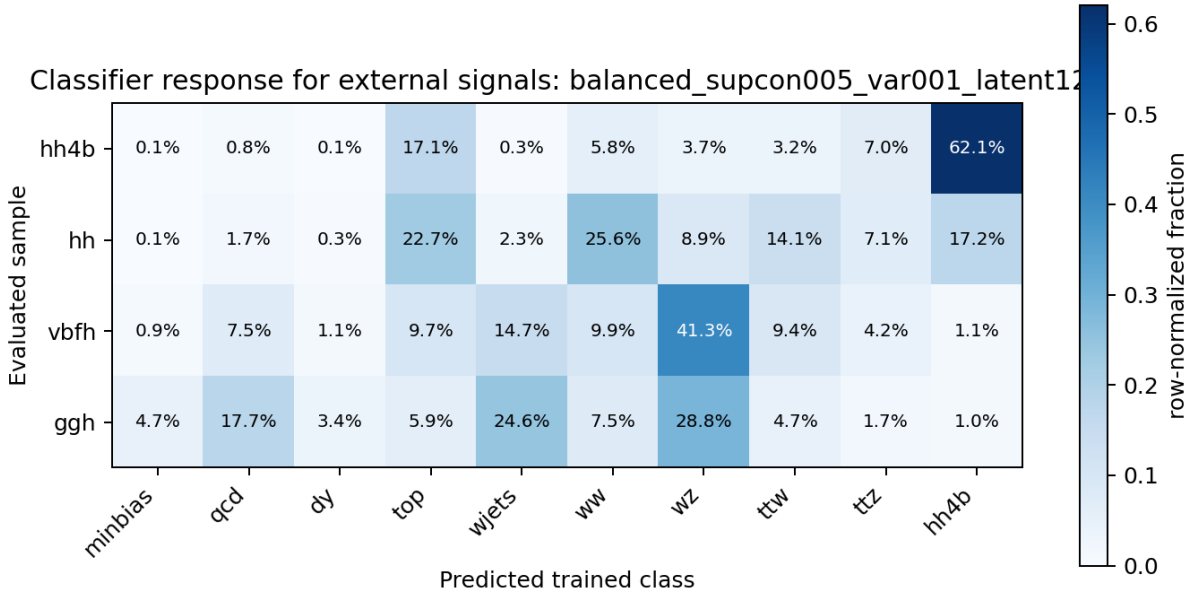
- This sweep was requested to check whether more latent capacity improves the representation.
- **Effective rank decreased** as latent_dim increased (5.05 → 4.45 → 1.51) — the opposite of the naive expectation.
- **eval-quality collapsed near zero at latent_dim=64** (0.00004), then recovered at 128 — a non-monotonic pattern similar to instabilities seen in earlier sweeps.
- HH4B F1 stayed essentially flat (0.559-0.564) across all three — capacity alone did not change anomaly-relevant classification.

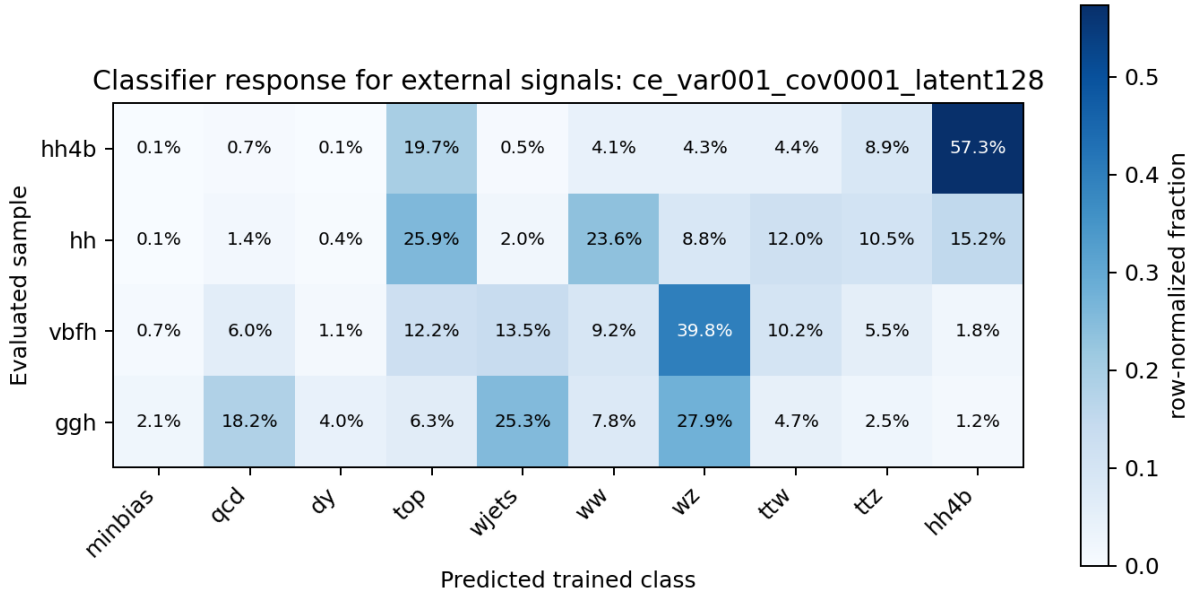
Testing signals the model has never seen in training



- External Higgs-like samples evaluated: HH, VBFH, ggH — plus trained HH4B as reference.
- These samples are evaluated as signals against the 9-class SM background (minbias, qcd, dy, top, wjets, ww, wz, ttw, ttz).
- For unseen samples, classifier response is not accuracy — there is no ground-truth label to score against. It shows how the model maps them into known trained classes.
- Both matter: anomaly score (latent-space geometry) and classifier response (class mapping) capture different things and can disagree.

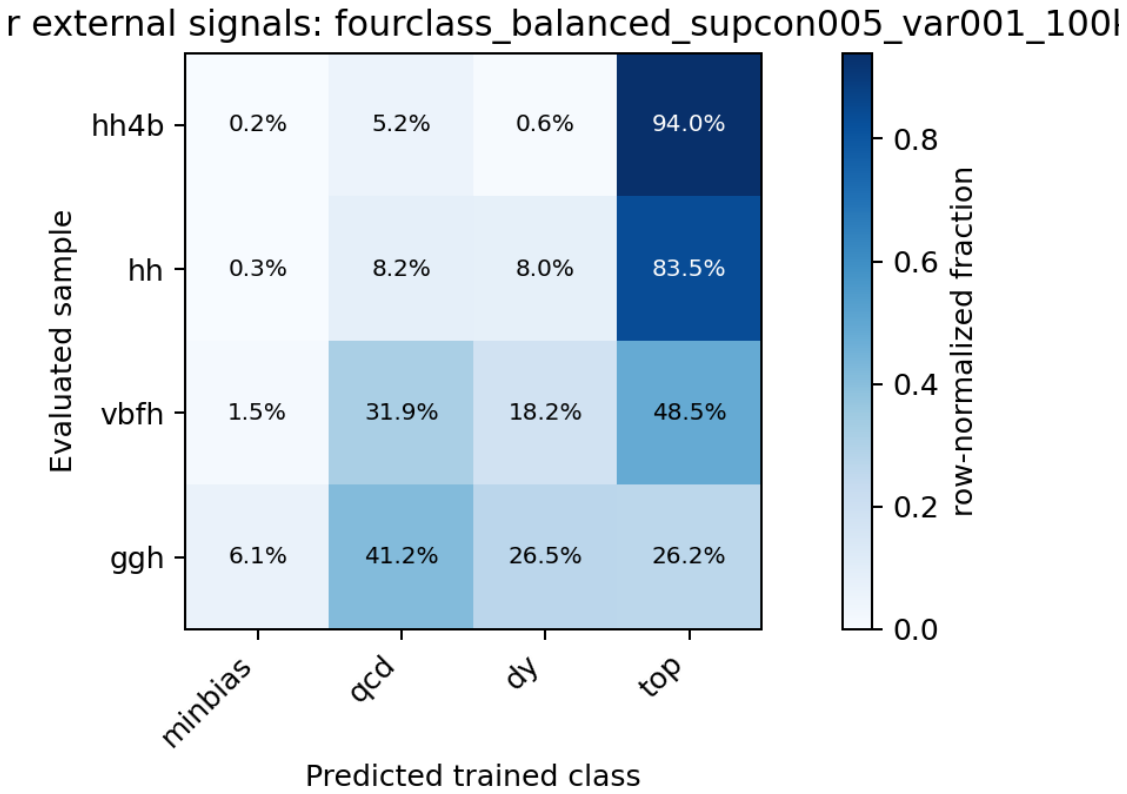
Balanced vs. anomaly-oriented objective





SIGNAL	TOP CLASS (BAL.)	HH4B TAG (BAL.)	TOP CLASS (ANOM.)	HH4B TAG (ANOM.)
hh4b (trained)	hh4b	0.6205	hh4b	0.5730
hh	ww	0.1725	top	0.1520
vbfh	wz	0.0115	wz	0.0180
ggh	wz	0.0100	wz	0.0125

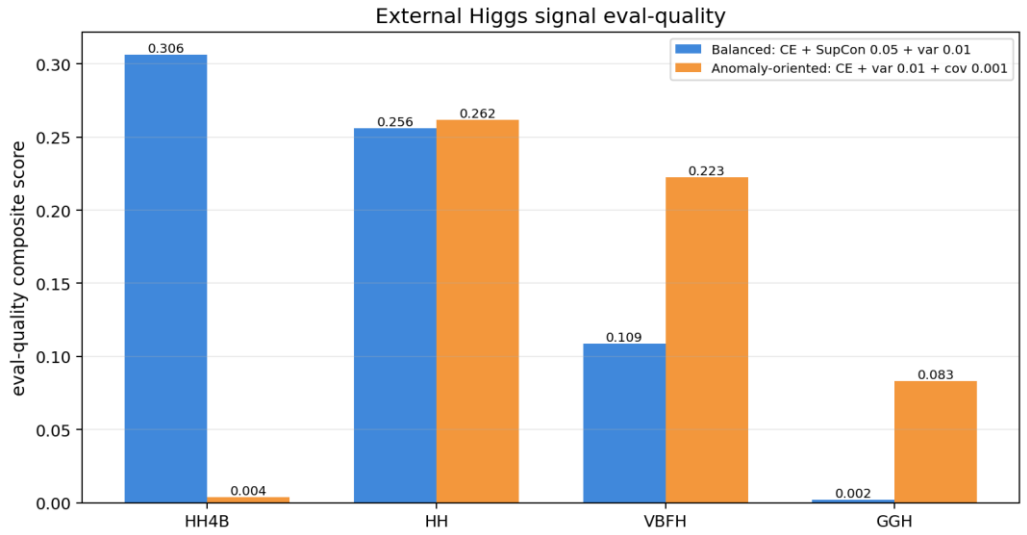
4-CLASS VS. 10-CLASS COMPARISON

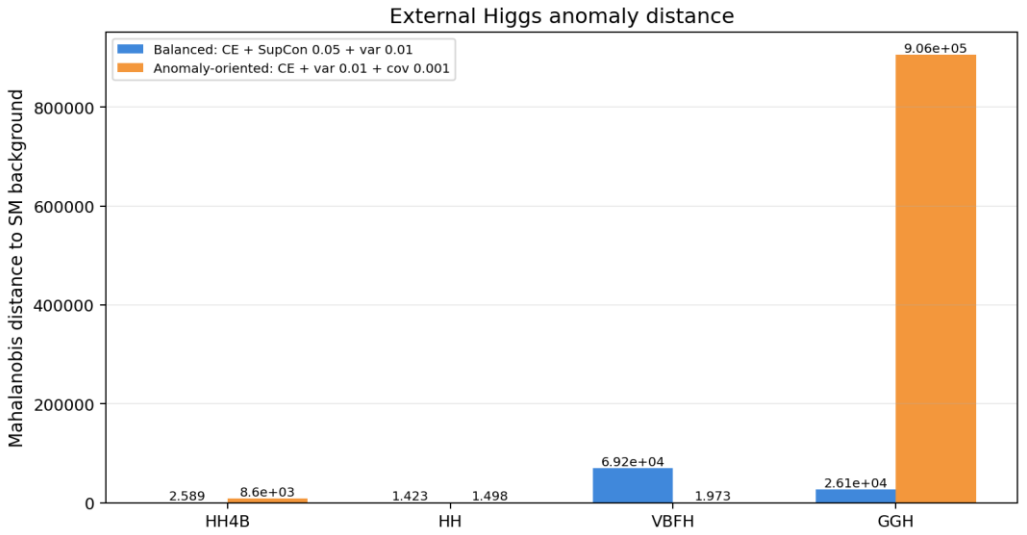


SIGNAL	TOP MAPPED CLASS	TOP FRACTION	HH4B TAG RATE	MEAN HH4B PROB.
hh4b	top	0.9400	*not available*	*not available*
hh	top	0.8350	*not available*	*not available*
vbfh	top	0.4850	*not available*	*not available*
ggh	qcd	0.4125	*not available*	*not available*

- The 4-class model is trained only on minbias, qcd, dy, top — it has no HH4B output node.
- HH4B, HH, and VBFH are all mostly mapped into Top; ggH mostly into QCD.
- **The 10-class model** can directly test whether HH-like signals activate an HH4B output; the 4-class model structurally cannot.

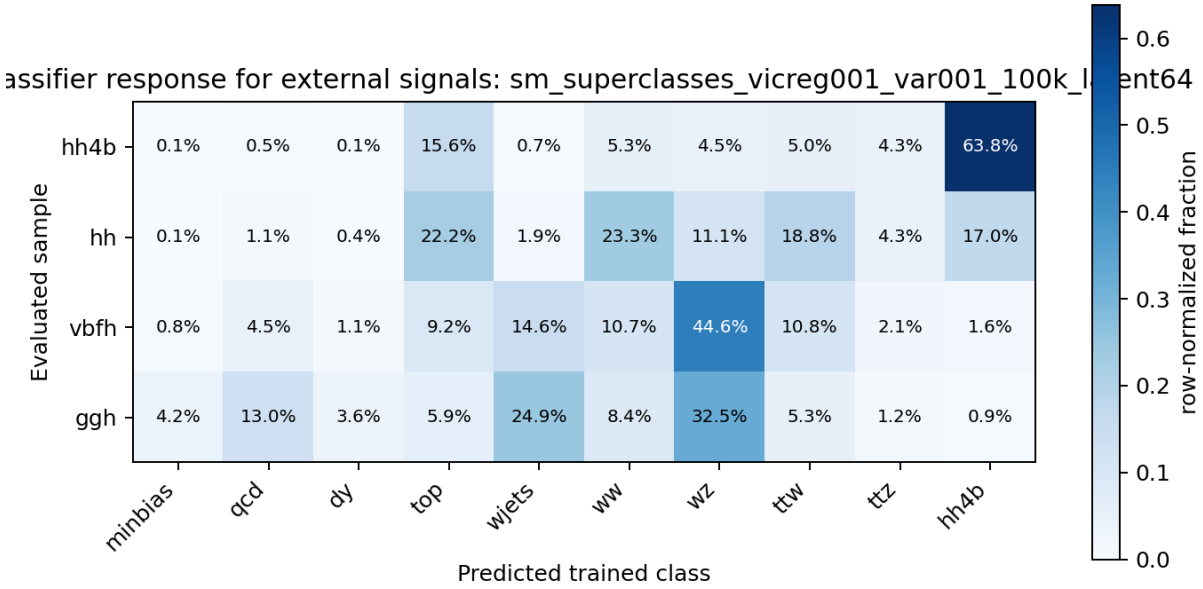
Classifier response and anomaly score can disagree





- The anomaly score is evaluated through latent-space geometry (Mahalanobis distance, GMM gap, composite eval-quality).
- **hh4b eval-quality**: 0.306 (balanced) vs. 0.0036 (anomaly-oriented) — despite its name, the anomaly-oriented model scores trained HH4B lower on this composite metric.
- **ggh Mahalanobis**: 9.06×10^5 under the anomaly-oriented model — far larger distance from background than any other signal/model pair.
- This disagreement is exactly why classifier response alone is not enough for an anomaly-detection project.

Promising for trained HH4B, not yet for external signals



METRIC	LATENT_DIM=64	LATENT_DIM=128
HH4B tag rate (trained)	0.6385	0.5915
HH tag rate (external)	0.1695	0.1505
VBFH tag rate (external)	0.0155	0.0125
ggH tag rate (external)	0.0095	0.0100
HH4B eval-quality (geometry)	0.1098	0.1702
Effective rank	7.23	6.31
Head-Fisher effective rank	61.05	89.22

- **VICReg latent_dim=64 gives the highest trained-HH4B tag rate of any model evaluated (0.6385)**, ahead of both the balanced and anomaly-oriented 10-class models.
- It does not yet clearly make HH, VBFH, or ggH map to HH4B — external tag rates remain low, similar to the other models.
- **Promising but not yet the main selected model – exploratory test**

Main findings

- The 10-class setup is more realistic for anomaly detection than the earlier 4-class setup — it can directly test whether HH-like signals activate an HH4B output.
- More granular/broader labels activate more latent structure than the original coarse setup (consistent with the effective-rank spread seen across objectives).
- **External Higgs signals do not all behave like trained HH4B at L1.**
- **HH** shows partial overlap with HH4B-like behavior (tag rate ~0.15-0.23 depending on model).
- **VBFH and ggH** are consistently mapped to the WZ electroweak-like background class under both 10-class models (tag rates ~1-2%).
- Balanced (CE + SupCon 0.05 + variance) remains the most stable main candidate across metrics.
- Covariance-regularized and VICReg-style models are useful anomaly-geometry comparisons, each strong on different metrics.

MAIN — *balanced* — **CE + SupCon 0.05 + variance**

COMPARISON — *anomaly-oriented* — **CE + variance + covariance**

EXPLORATORY — *not yet selected* — **VICReg-style**